

1 Set theory

As a consequence, we have that taking countable unions still cannot increase cardinality.

Theorem 1.68. *Let A_n be a countable set for each $n \in \mathbb{N}$. Then $\bigcup_n A_n$ is countable.*

Proof. Since A_n is countable, there is a surjection $f_n : \mathbb{N} \rightarrow A_n$. Define a function $f : \mathbb{N} \times \mathbb{N} \rightarrow \bigcup_n A_n$ by

$$f(m, n) = f_n(m).$$

This function is surjective, hence $\bigcup_n A_n$ is countable. □

How about the product of countably many countable sets?

Theorem 1.69. *$\{0, 1\}^{\mathbb{N}}$ is uncountable. So the countable product of countable sets need not be countable.*

Proof. Suppose $\{0, 1\}^{\mathbb{N}}$ were countable. Then we could list all infinite 0–1 sequences as

$$s^{(1)}, s^{(2)}, s^{(3)}, \dots, \quad s^{(k)} = (s_1^{(k)}, s_2^{(k)}, s_3^{(k)}, \dots), \quad s_n^{(k)} \in \{0, 1\}.$$

Define a new sequence $t = (t_1, t_2, \dots)$ by flipping the diagonal bits:

$$t_n = 1 - s_n^{(n)} \quad (n \geq 1).$$

Then $t \in \{0, 1\}^{\mathbb{N}}$, but $t \neq s^{(k)}$ for every k because they differ in the k -th coordinate. This contradicts the assumption that the list was complete. Hence $\{0, 1\}^{\mathbb{N}}$ is uncountable. □

Corollary 1.70. $|\mathcal{P}(\mathbb{N})| = |\{0, 1\}^{\mathbb{N}}| > |\mathbb{N}|$.

Proof. Define $\Phi : \{0, 1\}^{\mathbb{N}} \rightarrow \mathcal{P}(\mathbb{N})$ by

$$\Phi((x_n)_{n \geq 1}) = \{n \in \mathbb{N} : x_n = 1\}.$$

If $(x_n) \neq (y_n)$ pick k with $x_k \neq y_k$; then k belongs to exactly one of the two images, so $\Phi(x_n) \neq \Phi(y_n)$. Thus Φ is injective. Conversely, for any $A \subseteq \mathbb{N}$ let its characteristic sequence χ_A be given by $\chi_A(n) = 1$ if $n \in A$, 0 otherwise. Then $\Phi(\chi_A) = A$, so Φ is surjective. Hence $|\mathcal{P}(\mathbb{N})| = |\{0, 1\}^{\mathbb{N}}|$. $\square$

In fact, this corollary is a special case of a more general result which we will prove later.

Theorem 1.71 (Cantor's Theorem). *For any set S , there is no surjective function from S to its power set $\mathcal{P}(S)$. In particular, $|S| < |\mathcal{P}(S)|$.*

Corollary 1.72. $|[0, 1]| = |\{0, 1\}^{\mathbb{N}}| > \aleph_0$.

Proof. Define $E \subset \{0, 1\}^{\mathbb{N}}$ to be the set of sequences that are not eventually all 1. Define $\Phi : E \rightarrow [0, 1]$ by

$$\Phi((x_n)_{n \geq 1}) = \sum_{n=1}^{\infty} \frac{x_n}{2^n}.$$

This is surjective because for any $y \in [0, 1]$, we can write y in binary expansion $y = 0.y_1y_2y_3\ldots$ with $y_i \in \{0, 1\}$ and not ending with an infinite sequence of 1's (i.e., when $0.x_1\cdots x_{m-1}1000\cdots = 0.x_1\cdots x_{m-1}0111\cdots$ happens we choose the expansion that does not end with 1's).

This is also injective: Let $(x_k), (y_k) \in E$ with $(x_k) \neq (y_k)$ and let n be the first index where they differ. WLOG $x_n = 0$ and $y_n = 1$. Then

$$\Phi(y) - \Phi(x) = \sum_{k=1}^{\infty} \frac{y_k - x_k}{2^k} = \frac{1}{2^n} + \sum_{k>n} \frac{y_k - x_k}{2^k}.$$

Since $y_k - x_k \in \{-1, 0, 1\}$,

$$\sum_{k>n} \frac{y_k - x_k}{2^k} \geq -\sum_{k>n} \frac{1}{2^k} = -\frac{1}{2^n},$$

with equality only if $y_k - x_k = -1$ for all $k > n$, i.e. $x_k = 1$ and $y_k = 0$ for every $k > n$, which would make (x_k) eventually 1, contrary to $(x_k) \in E$. Hence the tail sum is strictly greater than $-1/2^n$, so $\Phi(y) - \Phi(x) > 0$ and therefore $\Phi(x) \neq \Phi(y)$.

Therefore, $|[0, 1]| = |E|$. Now, we let $D \subset \{0, 1\}^{\mathbb{N}}$ be the set of sequences that are eventually 1. $E = \{0, 1\}^{\mathbb{N}} \setminus D$. We use the following two claims to finish the proof.

Claim 1 D is countable.

Claim 2 An uncountable set minus a countable set will maintain its cardinality.

Hence, we have $|E| = |\{0, 1\}^{\mathbb{N}}|$. Hence $|[0, 1]| = |\{0, 1\}^{\mathbb{N}}| > \aleph_0$.

We will leave Claim 1 as an exercise in your homework 2. We prove Claim 2 below. $\square$

Proposition 1.73. *Let A be an infinite set and F a finite set. Then,*

- $|A \cup F| = |A|$.
- $|A \setminus F| = |A|$.

The conclusion still holds if A is uncountable and F is infinitely countable.

Proof. Let $G := F \setminus A$. Then G is finite and disjoint from A . We need to show that $|A \cup G| = |A|$. Since A is infinite, we can choose a sequence of distinct elements $a_1, a_2, \dots$ in A . If $G = \emptyset$, the result is trivial. If $G \neq \emptyset$, we write $G = \{g_1, g_2, \dots, g_n\}$, where n is finite. Define $f : A \rightarrow A \cup G$ as follows:

1. if $a \notin \{a_1, \dots\}$, then set $f(a) := a$;
2. if $a = a_i$ for some $i \leq n$, then set $f(a) := g_i$.
3. If $a = a_i$ for some $i > n$, then set $f(a) := a_{i-n}$.

Then, f is a bijection. Thus $|A \cup G| = |A|$.

For the second part, we let $B := A \setminus F$. Then, $|A \setminus F| = |B| = |B \cup (A \cap F)| = |A|$.

Now, assume that F is countably infinite and A is uncountable. Let $G := F \setminus A$. Then, G is countable. WLOG, we assume G is infinite. We then can write $G = \{g_1, g_2, \dots\}$. Still choose a sequence of distinct elements $a_1, a_2, \dots$ in A . Define $f : A \rightarrow A \cup G$ as follows:

1. if $a \notin \{a_1, \dots\}$, then set $f(a) := a$;
2. if $a = a_{2n}$, then set $f(a_{2n}) := g_n$;
3. if $a = a_{2n-1}$, then set $f(a_{2n-1}) := a_n$.

Then, f is a bijection. Thus $|A \cup G| = |A|$. $\square$

Corollary 1.74. $|[0, 1]| = |[0, 1]| = |(0, 1)| = |(0, 1]|$.

Proposition 1.75. $|(0, 1)| = |\mathbb{R}|$.

Proof. Let $f : (0, 1) \rightarrow \mathbb{R}$ be defined by

$$f(x) = \tan\left(\left(x - \frac{1}{2}\right)\pi\right).$$

Then f is a bijection, hence $|(0, 1)| = |\mathbb{R}|$. $\square$

Corollary 1.76. $|\mathbb{R} \setminus \mathbb{Q}| = |\mathbb{R}|$.

1.5.1 Proofs of some general theorems

Theorem 1.77 (Cantor-Schröder-Bernstein). *If there exist injections $f : A \rightarrow B$ and $g : B \rightarrow A$, then there exists a bijection $h : A \rightarrow B$.*

Proof. WLOG, we assume that A and B are disjoint (otherwise replace B by a disjoint copy). For each $a \in A$, we have the following sequence:

$$\dots \xrightarrow{f} b_{-3} \xrightarrow{g} a_{-2} \xrightarrow{f} b_{-1} \xrightarrow{g} a_0 = a \xrightarrow{f} b_0 \xrightarrow{g} a_1 \xrightarrow{f} b_1 \xrightarrow{g} a_2 \xrightarrow{f} \dots$$

There are only four possibilities for this sequence:

1. (B -stopper) it stops at some b_{-k} where $k \geq 1$.
2. (A -stopper) it stops at some a_{-k} where $k \geq 0$.
3. (bi-infinite) it continues indefinitely in both directions.
4. (cyclic) it is a cyclic sequence, i.e., $a_i = a_j$ for some $i < j$.

This also holds for any $b \in B$ and hence these four possibilities give rise to a partition of $A \sqcup B$.

Now, we define $h : A \rightarrow B$ as follows:

$$h(a) = \begin{cases} f(a), & \text{if } a \text{ is an } A\text{-stopper;} \\ f(a), & \text{if } a \text{ is in a bi-infinite chain or a cycle;} \\ g^{-1}(a), & \text{if } a \text{ is a } B\text{-stopper.} \end{cases}$$

This is a bijection. □

We proved this using Well-ordering theorem. We can also prove this using Zorn's lemma.

Theorem 1.78. *If A, B are sets then either $|A| \leq |B|$ or $|B| \leq |A|$.*

Proof. Let $\mathcal{M}$ be the set of *matchings* between A and B :

$$\mathcal{M} = \{M \subseteq A \times B \mid \text{each } a \in A \text{ appears in at most one pair of } M, \text{ and each } b \in B \text{ appears in at most one}\}.$$

We partially order $\mathcal{M}$ by inclusion. If $\{M_i\}_{i \in I}$ is a chain in $\mathcal{M}$, then

$$M = \bigcup_{i \in I} M_i$$

is again a matching, since no element of A or B can be matched to more than one partner within a chain. Thus every chain has an upper bound.

By *Zorn's Lemma*, $\mathcal{M}$ has a maximal element M^* .

Suppose there exist $a \in A$ and $b \in B$ both unmatched in M^* . Then

$$M' = M^* \cup \{(a, b)\}$$

is a matching strictly larger than M^* , contradicting maximality. Therefore, at least one side is fully matched: either every $a \in A$ is matched, or every $b \in B$ is matched.

- If every $a \in A$ is matched, define $f : A \rightarrow B$ by $f(a) = b$ whenever $(a, b) \in M^*$. This f is injective, so $|A| \leq |B|$.
- Otherwise, every $b \in B$ is matched. Define $g : B \rightarrow A$ by $g(b) = a$ whenever $(a, b) \in M^*$. Then g is injective, so $|B| \leq |A|$.

Thus, for any sets A and B , either $|A| \leq |B|$ or $|B| \leq |A|$. $\square$

Theorem 1.79 (Cantor's Theorem). *For any set S , there is no surjective function from S to its power set $\mathcal{P}(S)$. In particular, $|S| < |\mathcal{P}(S)|$.*

Proof. Suppose $f : S \rightarrow \mathcal{P}(S)$ is any function. Define $B = \{s \in S : s \notin f(s)\}$. If $B \subseteq S$ is in the range of f , then there exists $s_0 \in S$ such that $f(s_0) = B$. The following contradictions arise: one can show that

$$s_0 \in B \implies s_0 \notin f(s_0) = B \quad \text{and} \quad s_0 \notin B = f(s_0) \implies s_0 \in B,$$

which forces a contradiction. Consequently any function $f : S \rightarrow \mathcal{P}(S)$ must fail to be surjective, and hence $|S| \neq |\mathcal{P}(S)|$. On the other hand, there is an obvious injection $S \rightarrow \mathcal{P}(S)$ given by $s \mapsto \{s\}$. Therefore

$$|S| < |\mathcal{P}(S)| < |\mathcal{P}(\mathcal{P}(S))| < \dots$$

$\square$

2 Metric spaces

For real functions $f : \mathbb{R} \rightarrow \mathbb{R}$, we have the notion of continuity using ε - δ definition: a function $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous at a point $x_0 \in \mathbb{R}$ if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$|f(x) - f(x_0)| < \varepsilon \quad \text{whenever } |x - x_0| < \delta.$$

We also have the notion of continuity using sequences: a function $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous at a point $x_0 \in \mathbb{R}$ if for every sequence (x_n) in $\mathbb{R}$ converging to x_0 , the sequence $(f(x_n))$ converges to $f(x_0)$.

In an attempt to generalize these notions to more general settings than $\mathbb{R}$, Frechet introduced the notion of metric spaces in 1906. Metric spaces provide a natural setting to generalize the notions of continuity above. Later, Hausdorff introduced the more general notion of topological spaces in 1914. We will first study metric spaces and then topological spaces.

Definition 2.1 (Metric Space). A *metric space* is a pair (X, d) consisting of a set X and a function $d : X \times X \rightarrow [0, \infty)$ called a *metric* such that

1. $d(x, y) = 0$ if and only if $x = y$;
2. $d(x, y) = d(y, x)$ for all $x, y \in X$ (symmetry);
3. $d(x, z) \leq d(x, y) + d(y, z)$ for all $x, y, z \in X$ (triangle inequality).

The value $d(x, y)$ is called the *distance* between points x and y in the metric space.

If (X, d) satisfies 2 and 3 and 1' below, it is called a *pseudometric space*.

- 1'. $d(x, x) = 0$ for all $x \in X$.

Example 2.2. 1. $X = \mathbb{R}^n$, $d_p(x, y) = \|x - y\|_p = (\sum_{i=1}^n |x_i - y_i|^p)^{1/p}$ for $1 \leq p < \infty$ and $d_\infty(x, y) = \|x - y\|_\infty = \max_{1 \leq i \leq n} |x_i - y_i|$ are metrics on $\mathbb{R}^n$.

2. Let X be any set and $f : X \rightarrow \mathbb{R}$ be any function. Then, the following defines a pseudometric on X :

$$d(x, y) = |f(x) - f(y)|.$$

3. Let (X, d) be a metric space and $A \subset X$ be a nonempty subset. Then, $(A, d|_{A \times A})$ is a metric space, called a *subspace* of (X, d) .